Tank 8: Test and record

Goal: I can run the same test three times and record what actually happens.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 15, grade 6 path; Playbook B meeting 16, grade 6 path

Ask | Plan | Build | **Test** | Improve

You are done when

Show three results and identify one specific design problem.

You will use

- Prototype
- Approved test objects
- Paper
- Pencils
- Ruler or timer if needed

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Read today's goal.
- Look at the example.

2 Step 2: Tell your partner — 1 min

- Tell your partner one useful detail.

3 Step 3: Watch and try — 5 min

- Watch one trial from the same starting position.
- See a record include an unsuccessful result honestly.

4 Step 4: Make it work — 17 min

- Reset the test to the agreed starting position.
- Run three trials and record each result.
- Keep the design unchanged until the trials are finished.

5 Step 5: Check one thing — 5 min

- Compare all three results.
- Point to the most useful pattern or failure.
- Choose one problem to fix next.

6 Step 6: Reset together — 3 min

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

7 Step 7: Show your result — 2 min

- Show three results and identify one specific design problem.

What to turn in

Three recorded trials in the team folder.

No computer? Another way to do it

The teacher runs the test while the team records.

Finished early? Next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Stay safe

Use the approved test only. Stop if a part breaks sharply or the setup becomes unstable.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.